Causal Steering of Protein Language Models: When Correlational Validation Fails, and How to Catch It

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Abstract

Activation steering has recently been shown to reproduce a known protein property (thermostability) in ESM2-650M via difference-of-means vectors. We ask whether a scoring proxy's correlation with real experimental labels predicts whether steering toward it actually works, and find the answer is no – not weakly no. Under a pre-registered six-criterion protocol, we test five novel target properties and report the failures alongside the successes: binding affinity's proxy is the best-validated in our study ($r = 0.80$ held-out, confirmed by a weight-shuffle null control and held-out-position generalization) and steers nothing; a substantially weaker proxy ($r = 0.22$, catalytic activity) produces a robust, three-seed-replicated effect. Immunogenicity is killed at the proxy-validation gate before any model run, after a proxy that looks like a real signal ($r = 0.38$) turns out to be a source-organism confound. Expression yield passes the naive check but fails residue-exclusion; a follow-up diagnostic explains why: its steering vector is a partial geometric echo of a different, already-validated target's direction. This mirrors an independent, component-level finding in the same model family: attention heads ranked by correlation with real 3D contacts show near-zero relationship to their causal ablation effect. We further show a single-seed result can mislead in criterion-specific ways, and describe two near-misses caught in our own pipeline – a spurious pass from comparing arms at an unsafe intervention strength, and a robustness check that only holds on 2 of 3 seeds – as concrete evidence for why this level of scrutiny is necessary before any steering result is trusted. We release the full protocol, all target harnesses, and the seed-replication data.

1 Problem

Task and setting. Difference-of-means activation steering – adding a direction built from the mean activation difference between high- and low-property example groups to a protein language model's residual stream – has been shown to reproduce known thermostability biology in ESM2-650M [Huang et al., 2025]. The natural next question for anyone applying this technique to a new biological property is whether a scoring proxy's correlation with real experimental labels predicts whether steering toward it will work.

Assumptions being tested. The implicit assumption in adopting this technique for a new target is that (a) validating a compositional proxy against real labels is a reliable gate before running a steering experiment, and (b) a single steering run that clears a significance threshold is trustworthy evidence of a causal effect. We test five novel target properties – binding affinity, catalytic activity, intrinsic disorder, immunogenicity, and expression yield – under one pre-registered protocol (§2) designed specifically to stress-test both assumptions, and report every outcome, not only the successes.

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Table 1: Summary of all five novel target properties. Proxy r is Pearson correlation against real labels on a held-out split, computed before any steering run.

Target	Dataset	Proxy r	Steering result	Verdict
Binding affinity	ProteinGym DMS (RASK/DARPin)	0.80	flat null, all α	KILL
Catalytic activity	DLKcat (cross-protein)	0.22	PASS, 3/3 seeds	**PASS**
Intrinsic disorder	DisProt	0.45	PASS, 3/3 direction; 2/3 robustness	**PASS***
Immunogenicity	IEDB (4-tier)	$\leq$0.10	not run	KILL (pre-run)
Expression yield	eSol	0.31	fails residue-exclusion	AMBIGUOUS

*PASS on criteria 1, 2, 4, 6, and 2 of 3 seeds on criterion 3 (§3.1).

Target metrics. For each property: (1) Pearson r of the compositional proxy against real experimental labels on a held-out split, computed *before* any steering run; (2) the real steering direction's effect size against a matched-norm random-direction control, via paired bootstrap; (3) whether that effect survives a residue-exclusion robustness check; (4) whether the result replicates across independent random seeds. A "desired improvement" would be a positive, monotonic, seed-stable relationship between proxy validity and steering effect size – which we do not find.

2 Proposed Approach

We lock a six-criterion decision rule before testing any new target, computed automatically at the end of each run rather than eyeballed after the fact: (1) the real direction significantly beats a matched-norm random-direction control via paired bootstrap (10,000 resamples); (2) a coherent, monotonic dose-response across at least three steering strengths (α), restricted to a pre-established safe operating range (§4); (3) the effect survives residue-exclusion, ruling out pure compositional collapse; (4) the proxy is validated against real labels *before* the steering run; (5) the result beats the best existing technique where one exists; (6) at least 150 held-out evaluation sequences. A target failing any criterion outright is a **KILL**; one passing 1–2 but failing 3 or 4 is **AMBIGUOUS**.

The core mechanism under test in each of the five targets is identical: build a per-layer difference-of-means steering vector from real high/low-property sequence groups, add it to ESM2-650M's residual stream at generation time, and score the result with a compositional proxy validated in advance. The hypothesis under which this is expected to work is Huang et al.'s finding for thermostability [Huang et al., 2025] generalizing to other properties whenever the scoring proxy is itself well-validated. We also recap an independent, component-level precondition test: Vig et al. [Vig et al., 2021] found an attention head in a protein language model whose attention pattern strongly correlates with real 3D contacts; we ablate it directly and measure the actual causal effect, and extend this to every head in the model (480 total), to test whether correlational salience predicts causal necessity one level below the steering question.

3 Observed Outcome

The central negative result: proxy validity and steering effect size show no positive relationship (Table 1, Figure 1). Binding affinity's proxy is the best-validated of any target we tried ($r = 0.80$ held-out, confirmed by a weight-shuffle null control giving $r = -0.001$, held-out-position generalization at $r = 0.54$, and mutational-load extrapolation at $r = 0.70$) – and the steering effect is a flat, unambiguous null at every tested α, with zero degenerate sequences at any strength. Catalytic activity's proxy is the weakest of the four runnable targets ($r = 0.22$) and produces a clean, monotonic, residue-robust effect that replicates identically across three independent seeds (Figure 2).

Component-level precondition, same pattern. Vig et al.'s attention head [Vig et al., 2021] shows 12.9× contact enrichment over background, but ablating it produces an effect statistically indistinguishable from ablating a low-correlation control head. A full sweep of all 480 heads, ranked purely by measured causal effect with zero correlational information, shows the correlational top pick ranks

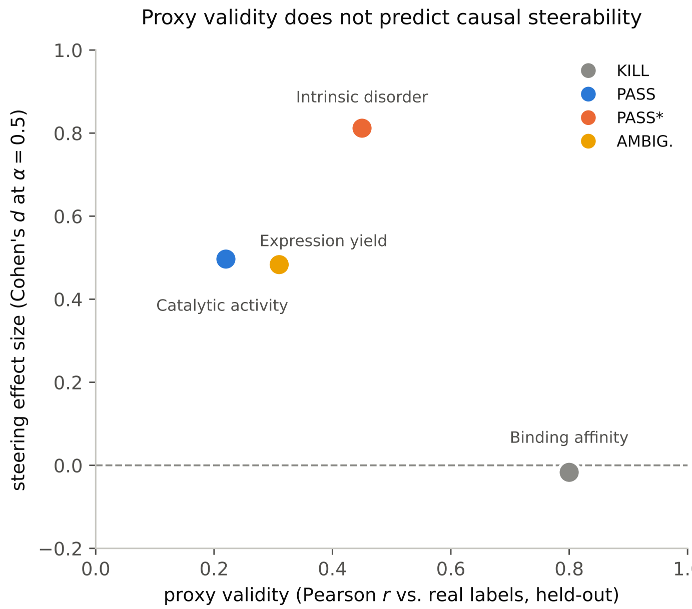

Figure 1: The four runnable targets from Table 1, plotted as proxy validity (Pearson r) against steering effect size (Cohen's d at $\alpha = 0.5$). The best-validated proxy (binding affinity, $r = 0.80$) produces the smallest effect (indistinguishable from zero); a proxy less than half as strong (catalytic activity, $r = 0.22$) produces one of the two largest effects. There is no visible positive trend.

313th of 480, and 166 of 480 heads (35%) show exactly zero measurable causal effect – confirming this is not an unlucky choice of control.

Immunogenicity: killed before any model run. Compositional proxies that look predictive on a surrogate binding-affinity tier ($r = 0.37$–0.43) collapse to near-chance on the real T-cell-response endpoint ($r \leq 0.10$), and a full-length-antigen proxy that appears to reach $r = 0.38$ is a source-organism confound (organism identity alone explains 58% of the apparent signal; the same proxy's correlation flips to -0.32 once organism is held out of cross-validation). No proxy survives criterion 4, so no steering script was written.

Expression yield: passes the naive check, fails the real one. The proxy validates at $r = 0.31$–0.34 and produces a significant, dose-responsive steering effect – which collapses to statistical insignificance under residue-exclusion, the same compositional-collapse shape our protocol exists to catch.

3.1 Intrinsic disorder: a seed-sensitive result, reported honestly

Using DisProt, we build a proxy from the published TOP-IDP disorder scale [Campen et al., 2008] ($r = 0.45$ held-out, the strongest proxy in our sweep, confirmed at the per-residue level with AUC 0.71). The steering effect is significant and monotonically dose-responsive on every one of three independent seeds (Figure 2). However, the residue-exclusion robustness check specifically passes on 2 of 3 seeds (36–42% of effect magnitude retained, CI excludes zero) and fails on the third (10% retained, CI crosses zero) – the effect's existence and direction are seed-stable, but the artifact check that rules out pure compositional collapse is not uniformly so (Figure 3).

4 Reason for Failure

Binding affinity's null is mechanistically explained, not just observed. The vector-building groups for this target are single-backbone point mutants (1–2 residues different from a shared 188-residue reference), giving the difference-of-means construction very little raw compositional signal to work with, independent of whether binding affinity is causally steerable at all. We measure this directly:

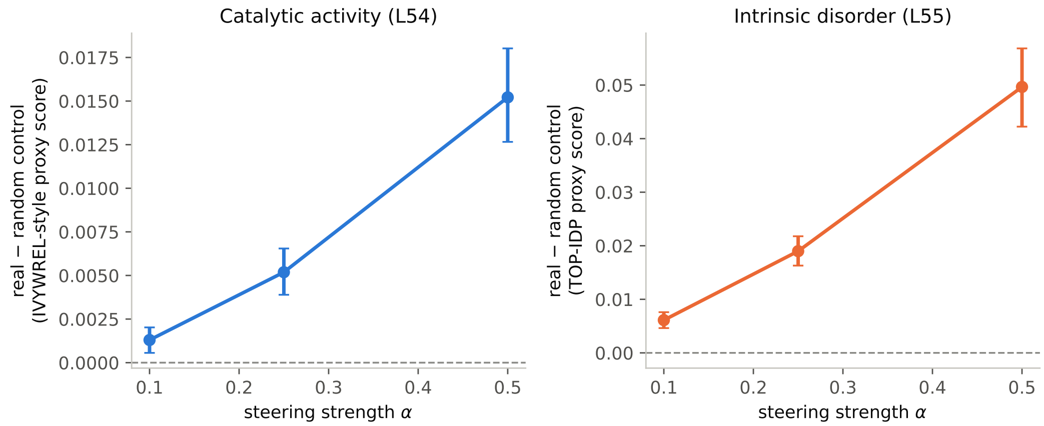

Figure 2: Dose-response for the two targets that pass criterion 2. Both show a clean, monotonic increase with 95% bootstrap CIs bounded away from zero at every point.

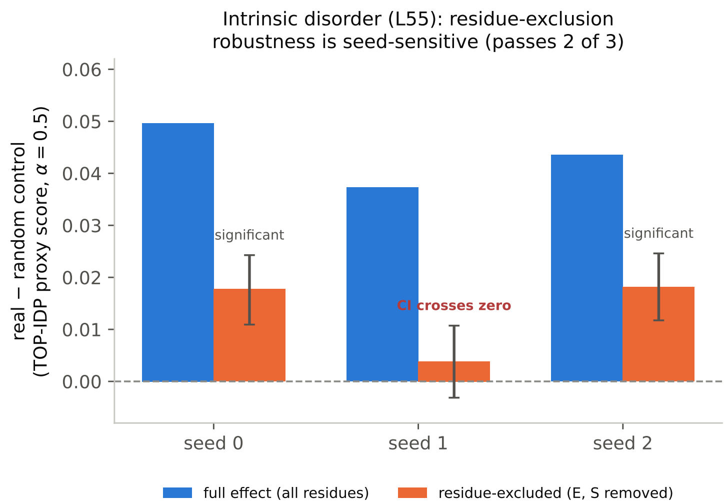

Figure 3: The intrinsic-disorder target's effect (blue) and the same effect after the two dominant substituted residues are excluded (orange, 95% bootstrap CIs), across three independent seeds. The full effect is stable across all three; the excluded-residue effect is significant on seeds 0 and 2 but its CI crosses zero on seed 1 – the same underlying effect, the same criterion, three different verdicts.

mean per-residue compositional distance between the low/high vector-building groups is 0.003 for this target, versus 0.02–0.05 for the three cross-protein datasets in our sweep. A well-validated scoring function does not guarantee the underlying difference-of-means construction has anything to act on.

Expression yield's failure is explained by vector geometry, not discarded as noise. We compute cosine similarity between this target's steering vector and every other target's, layer by layer, and find substantial overlap with the intrinsic-disorder target's vector: +0.38 on the full 33-layer concatenation, rising to +0.56–0.67 at the three deepest layers. This is consistent with the independently documented biology (disordered regions reduce solubility and expression yield): expression yield's apparent signal is best understood as a partial, noisier echo of disorder's already-validated real direction, filtered through a proxy more prone to compositional collapse than TOP-IDP – not independent evidence of a sixth steerable property.

A near-miss that would have hidden the real boundary condition. As a mechanistic follow-up on thermostability, we tested whether restricting steering to the 5 layers previously identified as causally necessary preserves the effect of steering all 33. A first implementation selected its best-performing α from the full tested range, including $\alpha \geq 1.0$, and reported a clean pass: the 5-layer subset appeared to outperform the full intervention at $\alpha = 2.0$. This was an artifact of comparison: at that α, the 33-layer intervention had already fully collapsed into degenerate output (zero non-degenerate evaluation sequences), while the 5-layer subset had not yet collapsed – it "won" only because it broke down later, not because it steered harder in any comparable regime. Restricting the comparison to a pre-established safe operating range and rerunning end to end gives the real result: the 5-layer subset produces a real, residue-robust effect, but retains only 43–59% of the full effect size – informative, not the unqualified success the uncollapsed comparison suggested.

Boundary conditions and actionable takeaways. (1) Proxy validation strength is not a reliable filter for which targets are worth attempting – check the raw compositional separation between vector-building groups, not just the scoring function's correlation with ground truth. (2) Any α-selection or best-config comparison must be restricted to a range where every compared arm is known non-degenerate, or a collapse-order artifact can produce a spurious win. (3) A criterion-specific robustness check (e.g., residue-exclusion) can have different seed-sensitivity than the headline effect itself, and needs its own multi-seed check, not inheritance from the main effect's stability. (4) An AMBIGUOUS or unexplained result is worth a cheap vector-geometry cross-check against other validated targets in the same study before being written off as pure noise.

5 Related Work

Huang et al. [2025] introduce difference-of-means activation steering for protein language models and validate it on thermostability; we extend their harness to five new properties under a pre-registered protocol and test, for the first time to our knowledge, whether proxy correlation strength predicts steering success across multiple unrelated properties. Vig et al. [2021] identify the attention head we redo as a genuine causal test and extend to a full 480-head sweep. Adams et al. [2025] train sparse autoencoders on ESM-2 and show via linear probing – a correlational method – that known determinants of thermostability and localization are identifiable in SAE feature space; their one steering experiment is a coherence check, not a validation against a real label, and their own discussion states that "investigating how to steer the model in useful ways remains an open direction for future work." We answer a version of that question for a non-SAE steering mechanism, specifically for whether correlational validity predicts causal usefulness.

6 Limitations

All experiments use a single model (ESM2-650M) at a single scale. All proxies are compositional heuristics chosen for interpretability and speed, not learned predictors – a stronger, non-compositional proxy might behave differently. Each target's final proxy was selected as the best of several candidates (typically 7–9) scored on the same held-out split whose r is then reported as the validation number, rather than via a separate train/select/test split – this garden-of-forking-paths bias optimistically inflates every reported proxy-r in the same direction, so the absolute values in Table 1 should be read as upper bounds; it does not obviously overturn the central ordinal comparison (binding affinity remains best-validated and null). We ran a large number of significance tests overall (5 targets $\times$ 5 alphas, 3-seed reruns for two targets, and the 480-head sweep) without a multiple-comparisons correction, partially mitigated by requiring multi-alpha and multi-seed agreement for any PASS, but borderline criteria (e.g. §3.1's residue-exclusion check, significant on only 2 of 3 seeds) are close enough to the noise floor that this remains a real caveat. We tested seed-robustness for two targets (catalytic activity, intrinsic disorder) but not all five. We did not run any structural or functional plausibility check (e.g. predicted-fold confidence) on generated sequences, so a PASS does not rule out that steered sequences, while scoring well on the proxy, are otherwise implausible proteins. Every steering result here is causal about the model's activation space – a real intervention evaluated against a random-direction control and residue-exclusion check – not validated against a real biological assay; a PASS means "causally steers a validated compositional proxy," not "causally engineers the underlying biological property."

LLM usage disclosure

Large language model assistance (Claude, Anthropic) was used throughout this project for code authoring, experiment scripting, literature lookup, and drafting of this manuscript. All experimental design decisions, the pre-registered protocol, and the interpretation of results were made and verified by the authors; every quantitative claim in this paper is backed by a runnable script and a committed results file, checked against the paper's prose during preparation.

Code and data availability

Code and results supporting every number in this paper are publicly available. An anonymized mirror preserving reviewer double-blindness will be linked in the final submission; the de-anonymized repository is released regardless of outcome.

Reproducibility statement

All five target properties in Table 1 reuse one fixed harness (steering-vector construction, matched-norm random control, paired bootstrap significance test, degeneracy filter), varying only the target dataset and scoring proxy, so a single implementation bug cannot selectively explain a subset of the reported results. Every quantitative claim – including both near-misses in §4 – is backed by a committed, re-runnable script and a saved results file; the two multi-seed claims (catalytic activity, intrinsic disorder) rerun the identical pipeline end to end per seed rather than resampling a single run's output.

References

Etowah Adams, Liam Bai, Minji Lee, Yiyang Yu, and Mohammed AlQuraishi. From mechanistic interpretability to mechanistic biology: Training, evaluating, and interpreting sparse autoencoders on protein language models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 460–476, 2025.

Andrew Campen, Ryan M. Williams, Celeste J. Brown, Jingwei Meng, Vladimir N. Uversky, and A. Keith Dunker. Top-idp-scale: a new amino acid scale measuring propensity for intrinsic disorder. *Protein and Peptide Letters*, 15(9):956–963, 2008.

Long-Kai Huang, Rongyi Zhu, Bing He, and Jianhua Yao. Steering protein language models. *arXiv preprint arXiv:2509.07983*, 2025. Accepted to ICML 2025.

Jesse Vig, Ali Madani, Lav R. Varshney, Caiming Xiong, Richard Socher, and Nazneen Fatema Rajani. Bertology meets biology: Interpreting attention in protein language models. In *International Conference on Learning Representations*, 2021.